

Matrices:

$$R \rightarrow \begin{matrix} A. \\ \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix} \times \begin{matrix} B. \\ \begin{bmatrix} 0 & 2 & 1 & 0 & 0 \\ 2 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} \end{matrix} = D$$

$$\frac{0 \cdot 0 + 0 \cdot 2 + 1 \cdot 1 + 0 \cdot 0 + 0 \cdot 0}{R \cdot C} = 2 \quad \frac{0 \cdot 2 + 1 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0}{R \cdot C} = 0 \quad \frac{1 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0}{R \cdot C} = 1$$

$$D_{1,1} = \sum_{j=1}^5 A_{1,j} \cdot B_{j,1}$$

Kulkarni's Equations:

$$\hat{y}_i(X) \approx \hat{y} \quad z_i = w_1 x_1 + w_2 x_2 + b$$

$$a = \sigma(z) = \frac{1}{1 + e^{-z}} \quad q_i = \sigma(z_i)$$

$$a = \text{prediction??} \quad z_2 = w_3 q_1$$

$$\hat{y} = \sigma(z_2)$$



Transformation

Loss Functions:

Cross-Entropy:

$$L(y, \hat{y}) = -[y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$$

MSE:

$$L_{MSE} = \frac{1}{2} (y - \hat{y})^2$$

Softmax:

$$\hat{y}_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Categorical Cross Entropy:

$$L = - \sum_i y_i \log(\hat{y}_i)$$

Hinge Loss:

$$L = \max(0, 1 - y \hat{y})$$

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{k=1}^K e^{z_k}} \quad (N \times M)$$

upper limit
index < lower limit

Backpropagation:

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial w}$$

Loss also equals the gradient that makes us make sense!!

$$w = w - \eta \frac{\partial L}{\partial w}$$

$$w_{n+1} = w_n - \eta \frac{\partial L}{\partial w}$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z} (x_1 + x_2)$$

Feedforward Networks

$$a' = \sigma(w'x + b')$$

$$a'' = \sigma(w''a' + b'')$$

$$y = \sigma(w''a'' + b'')$$

$$x \rightarrow 0 \rightarrow 0 \rightarrow 0 \rightarrow y$$

$$\begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix} = x$$

$$w' = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$w'' = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$g(y) = 2x^3 - 5x + 4 = 6x^2 - 5 + 0$$

$$g'(y) = 6x^2 - 5 + 0$$

$$g''(y) = 12x$$

$$g'''(y) = 12$$

$$g^{(4)}(y) = 0$$

$$g^{(5)}(y) = 0$$

$$g^{(6)}(y) = 0$$

$$g^{(7)}(y) = 0$$

$$g^{(8)}(y) = 0$$

$$g^{(9)}(y) = 0$$

$$g^{(10)}(y) = 0$$

$$g^{(11)}(y) = 0$$

$$g^{(12)}(y) = 0$$

$$g^{(13)}(y) = 0$$

$$g^{(14)}(y) = 0$$

$$g^{(15)}(y) = 0$$

$$g^{(16)}(y) = 0$$

$$g^{(17)}(y) = 0$$

$$g^{(18)}(y) = 0$$

$$g^{(19)}(y) = 0$$

$$g^{(20)}(y) = 0$$

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$$g^{(25)}(y) = 0$$

$$g^{(26)}(y) = 0$$

$$g^{(27)}(y) = 0$$

$$g^{(28)}(y) = 0$$

$$g^{(29)}(y) = 0$$

$$g^{(30)}(y) = 0$$

$$g^{(31)}(y) = 0$$

$$g^{(32)}(y) = 0$$

$$g^{(33)}(y) = 0$$

$$g^{(34)}(y) = 0$$

$$g^{(35)}(y) = 0$$

$$g^{(36)}(y) = 0$$

$$g^{(37)}(y) = 0$$

$$g^{(38)}(y) = 0$$

$$g^{(39)}(y) = 0$$

$$g^{(40)}(y) = 0$$

$$L = \sum_{i=1}^N y_i \log(\hat{y}_i)$$

$$\partial L = y_i \cdot \frac{1}{\hat{y}_i}$$

$$w = w - \eta \frac{\partial L}{\partial w}$$

$$\eta = \text{learning rate}$$

$$b = \text{derivative (aka db)}$$

$$\text{Power rule } (x^n)' = nx^{n-1}$$

$$\text{Constant multiple } = c \cdot f(x) = c \cdot f'(x)$$

$$\text{Ex: } (8x^9)' = 8 \cdot (4x^8) = 72x^8$$

$$(x^3)' = 3x^{3-1} = 3x^2$$

$$g(y) = 2x^3 - 5x + 4 = 6x^2 - 5 + 0$$

$$g'(y) = 6x^2 - 5 + 0$$

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